

Survey on Methodology of Intrusion Detection in Industrial Control System Based on Artificial Intelligence

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Abstract—With the increasingly close connection between the industrial control system and the Internet, the industrial control system are facing more and more cyber threats. Intrusion detection technology is one of the important technologies to protect the security of industrial control systems. With the development of artificial intelligence, many researchers have applied artificial intelligence algorithms based on machine learning and deep learning in the industrial control intrusion detection systems. Compared with the traditional methods, the detection accuracy based on machine learning is higher, but the procedure of manual feature extraction limits its practical application. The detection method based on deep learning has a strong feature extraction ability, but its computational complexity is high. In many cases, the detection effect is not ideal in industrial control scenes with real-time requirements. Based on the analysis of the related research, this paper not only summarizes the current research status of intrusion detection in industrial control system based on artificial intelligence, but also deeply discusses the challenges faced in this field. It is pointed out that the intrusion detection algorithm for the industrial control system should be further improved according to the characteristics of the industrial control system, such as real-time requirement, limited computing resources, high data dimension, and high data noise.

Keywords—industrial control system, intrusion detection, artificial intelligence, machine learning, deep learning

I. INTRODUCTION

Industrial Control System (ICS) is composed of a series of automation components and networks, which can not only automatically control the equipment in the production process, but also collect data and monitor the production information [1]. At present, ICS is widely used in various industries. Some ICSs related to key industries such as electric power, water

conservancy, nuclear power, chemical industry and energy, are even the key production facilities of the country. Once being attacked, they will not only cause a huge loss to the property of these industries, but also pose a threat to the national security [3]. Traditional ICS is isolated from the Internet, and its security measures mainly focus on functional security. Generally, physical isolation is adopted as its security measures, and there are no security defense measures against various Internet attacks [4]. In recent years, with the rapid development of information and intelligence in various fields, the connection between ICS and the Internet is also increasing, which gives criminals the opportunity to explore and attack industrial control equipment. Therefore, the security problems ICSs are facing are increasingly prominent [5].

In recent years, cyber-attacks against ICS have emerged one after another. The famous attacks, such as Stuxnet in 2010 [6], "Flame" in 2012 [7] and viruses such as "Havex" in 2014, have brought huge economic losses to the attacked enterprises. What is more, some cyber-attacks can even affect social stability and threaten national security. For example, in 2015, Ukraine's power grid was extensively paralyzed due to a virus attack, which affected the lives of 1.5 million Ukrainian people [8]. In May 2017, thousands of enterprises in nearly 100 countries around the world were attacked by WannaCry ransomware, which affected key areas such as energy, transportation and communication in many countries [9]. In 2018, TSMC was attacked by WannaCry variant virus, which forced several chip manufacturing plants of TSMC to shut down. In 2020, British company Darktrace discovered a persistent virus attack by detecting the operating environment of international airports [10]. In April of the same year, an energy company in Portugal was attacked by ransomware Ryuk, resulting in the theft of a large number of confidential data of the company [11]. It can be seen that once ICS is

attacked, it will not only cause great economic losses, but also endanger national security. Therefore, it is essential to maintain the safety of ICS.

Intrusion detection system (IDS) is one of the important technologies to ensure the security of ICS. Intrusion detection system can monitor the network data in ICS in real time, extract and analyze the data features, and identify the security threats in industrial control network through effective detection algorithms. In recent years, with the proposal and popularization of 5G technology, ICS security has been paid more and more attention by the state, and many intrusion detection technologies for ICS have emerged in academic circles. The methods used in these technologies are different, and the emphasis of research is also different. At present, the detection algorithm based on artificial intelligence is widely used in the field of industrial control. The detection method based on artificial intelligence has high detection accuracy and strong generalization ability, and has become a research hot spot in the field of industrial control intrusion detection. In this paper, the intrusion detection methods of industrial control system based on artificial intelligence algorithm in recent years are studied, the problems and challenges faced by the current intrusion detection methods of industrial control are pointed out, and the future research direction is discussed in depth.

II. CHARACTERISTICS OF INDUSTRIAL CONTROL INTRUSION DETECTION

A. Network Structure of Industrial Control System

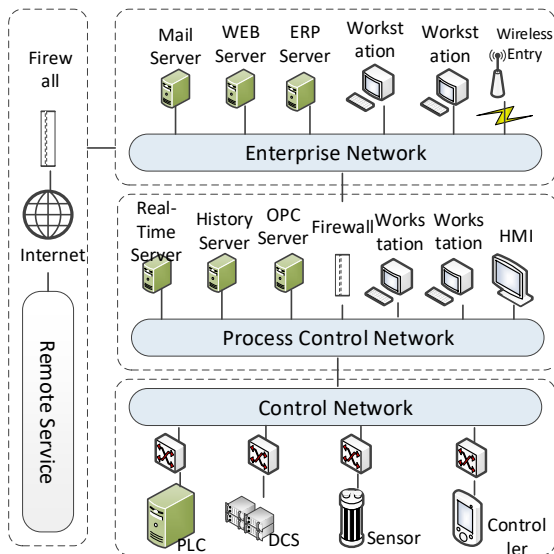


Fig. 1. Architecture diagram of industrial control system.

Generally, ICS is mainly composed of three network layers, as shown in Figure 1. From top to bottom, they are enterprise information network, process control network and field control network, and each layer of network corresponds to different functions. Enterprise information network is mainly used to supervise and control the industrial production process, which can facilitate managers to master the running state of the system and the running situation of equipment. The process control network mainly includes SCADA and data server,

which is responsible for transmitting the data of the underlying field control network to the enterprise information network. The field control network includes all kinds of field devices, such as PLC (Programmable Logic Controller) and RUT (Remote Terminal Unit), which are used to operate and sense the production equipment.

B. Characteristics of Industrial Control System Structure of Industrial Control System

- Industrial Internet is a network used between industrial control devices. At first, physical isolation was used to ensure the security of the system. Different from the traditional Internet, the industrial control Internet has its own uniqueness, which is mainly reflected in:
- Industrial Internet control is limited by the real-time requirement of industrial control system, and its primary goal is to ensure the communication efficiency between industrial control devices, not the communication security. Therefore, most communication protocols of industrial control networks lack security measures, which makes industrial control networks vulnerable to hacker attacks.
- In the same industrial scene, the types and quantities of industrial control equipment and sensors used in different industrial scales are different, and the network structures built are also different. This makes the attack on the same industrial scene take many forms. In the traditional Internet, the attacks on the same scene are similar.
- Paper[3] summarizes the characteristics of industrial control Internet. The uniqueness of industrial control Internet is mainly reflected in the following aspects: limited computing resources, real-time requirements, high data dimensionality, too much noise data, difficult to update and restart, etc.

C. Characteristics of the IDS for Industrial Control System

Because of these differences between industrial control internet and traditional internet, the intrusion detection of industrial control system is quite different from that of traditional internet [12], mainly reflected in:

- With the feature of real-time requirement in ICSs, it requires that the intrusion detection algorithm is able to process the real-time data of industrial control equipment.
- The computing and storage resources of ICS are very limited, which requires the intrusion detection algorithm for ICS with low resource consumption.
- ICS controls the operation of industrial control equipment, and it will not be suspended or restarted unless there is a critical condition. This requires the intrusion detection algorithm for ICS to have a high precision rate, and would rather miss the report than give a false alarm.

- In order to ensure the normal operation of industrial control equipment, the intrusion detection model in ICS should be updated as little as possible, which requires the intrusion detection model to have better generalization, that is, the model trained on the training set can be well applied in various real scenes.

III. INTRUSION DETECTION METHOD BASED ON MACHINE LEARNING

Early industrial control intrusion detection usually adopts the methods based on pattern matching [13] and the ones based on model. However, the methods based on pattern matching are difficult to detect the potential intrusion behavior that does not conform to the protocol specification, while the methods based on model can judge the intrusion behavior by observing the system parameters, and its timeliness is poor. In order to overcome the above disadvantages, many researchers began to study industrial control intrusion detection based on artificial intelligence algorithm of machine learning. Industrial control intrusion detection based on machine learning mainly includes several stages: data collection, feature extraction, behavior analysis and intrusion response. The specific process is shown in Figure 2.

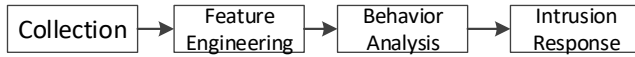


Fig. 2. Intrusion detection model based on ML.

In this paper, the industrial control intrusion detection algorithm based on machine learning is deeply studied, and it is classified into three learning methods: supervised, unsupervised and semi-supervised. The detailed introduction is shown in Table 1.

A. Intrusion Detection Algorithm Based on Supervised Learning

The supervised algorithm needs to spend a lot of energy on data labeling, but the supervised algorithm has a higher detection accuracy. In the field of intrusion detection, the commonly used supervised machine learning algorithms are NB(Naive Bayes), BN(Bayesian Network), HMM(Hidden Markov Models), KNN(K-Nearest Neighbor), DT(Decision Tree), SVM(Support Vector Machine) and LR(Logistic Regression)^[14-30]. Among them, the NB-based method is simple in logic and easy to implement^[14-16]; BN-based method has strong reasoning ability^[17,18]; HMM-based methods are good at dealing with problems with sequential nature^[19]; KNN algorithm does not need parameter estimation when solving multi-classification problems^[20-22]; DT algorithm is low in complexity and the construction rules are easy to understand^[23-25]; SVM is suitable for processing small sample data, and it also has strong generalization ability^[26,28]; LR is simple and fast in calculation [29,30]. Siphesihle et al. [112] evaluated several classifiers based on machine learning.

According to the characteristics of industrial control system data, the main supervised machine learning algorithm commonly used in industrial control intrusion detection include SVM^[31], BN^[35], HMM^[37] and KNN^[41] et al. SVM-based methods are similar to OCSVM-based methods, and they can't distinguish attack categories^[43]. A multi-classification

algorithm based on multiple SVM is proposed to detect the attack types in ICS. HMM algorithm is good at processing sequence data, and can mine the sequence rules and characteristics of industrial control data, and predict the data. Paper[39] proposed an intrusion detection model with two subsystems is proposed based on HMM. In this model, each subsystem contains several HMM classifiers, and when one of the classifiers detects an abnormality, the detection system will give an alarm. Although this detection method can improve the detection rate, it also leads to a high false alarm rate.

In addition, there are the industrial control intrusion detection algorithms based on the LR-based^[44], DT^[42], Boolean rule^[45] and Random Forests^[107].

B. Intrusion Detection Algorithm Based on Unsupervised Learning

In some industrial scenes, it is extremely difficult to obtain labeled data, so we can study intrusion detection technology through unsupervised algorithms. In the field of intrusion detection, the commonly used unsupervised machine learning algorithms mainly include k-means, hierarchical clustering and GMM (Gaussian Mixture Model). K-means algorithm has strong interpretability and fast convergence speed. In intrusion detection systems, K-means algorithm is usually combined with other algorithms. For example, in reference [59], the improved k-means algorithm is combined with SVM and ELM (extreme learning machine) to build an intrusion detection system, which can effectively detect Dos attacks. In Paper[60], k-means algorithm is combined with regression tree algorithm. Tomography clustering algorithm can find the hierarchical relationship between classes, and does not need to set the number of clusters in advance. In all references [61-64], hierarchical clustering is used to build intrusion detection models. GMM model can effectively identify abnormal behaviors similar to normal behavior distribution. In reference [65], GMM was used to model the behavior of normal data in NSL-KDD data set, and a classifier was trained to evaluate the probability of sample features, and based on this, abnormal data was identified. In reference [66], a distributed GMM model is proposed, which is based on Hadoop framework and can effectively reduce the computing time.

The data in ICS has the characteristics of limited resources and high real-time requirements. At present, the unsupervised learning methods commonly used in industrial control intrusion detection model based on machine learning are OCSVM (one-class support vector machine)^[69], K-means algorithm^[67] and Clustering algorithm. There are far more normal data than abnormal data in ICS data set, and the data set has great imbalance. However, OCSVM algorithm does not need to label data, but it can only be based on one kind of data. However, the detection model based on OCSVM algorithm generally has a high false alarm rate, and can't detect specific attack types.

Compared with OCSVM algorithm, the intrusion detection algorithm based on k-means can not only detect specific types of attacks, but also has higher performance. Paper[75]. This paper explores the detection effects of different clustering algorithms in industrial control intrusion detection model, and proves that K-means clustering algorithm can achieve better detection effects in industrial control intrusion detection. Many

researchers have studied industrial control intrusion detection based on K-means algorithm. For example, literature^[76]. In this paper, an unsupervised intrusion detection algorithm based on K-means is proposed. According to the characteristics of power system data, the algorithm clusters and analyzes the detection characteristic parameters. In this paper, a clustering algorithm based on fuzzy K-means is proposed. The algorithm deeply mines the data through multi-layer difference network, and can

classify and calculate the intrusion detection data of different levels. Paper[68]Based on K-means algorithm, an unsupervised industrial control intrusion detection algorithm is proposed. Clustering algorithm is highly dependent on researchers' understanding of data and their choice of features. When there are many noisy data in the data set, the clustering algorithm is mostly unsatisfactory.

TABLE I. SUMMARY OF THE INTRUSION DETECTION METHOD FOR INDUSTRIAL CONTROL SYSTEMS BASED ON ML

Papers	Machine learning method	evaluating indicator	data set	Supervision type
Paper[31]	SVM	Accuracy, detection rate, false alarm rate and stability	NSL-KD99 , UNSW-NB	Supervised
Paper[32]	SVM	Reactor pressure change	Unpublished data set	Supervised
Paper[33]	SVM	Detection rate, RMSE	SWaT	Supervised
Paper[34]	SVM	Accuracy, detection rate, false alarm rate, training speed		Supervised
Paper[35]	Bayesian network	Accuracy, running time	SWaT	Supervised
Paper[37]	HMM	Detection rate, detection time, false alarm rate and accuracy rate	Unpublished data set	
Paper[38]	Markov algorithm	Detection rate, false alarm rate	Semantic attack	Supervised
Paper[39]	HMM	Detection accuracy	Natural gas pipeline data, power transmission data, water tower data	Supervised
Paper[67]	K-means	Accuracy, recall, accuracy, false alarm rate	UCI data set	Supervised
Paper[41]	KNN	Accuracy, detection rate, recall rate, accuracy, F1 value, sensitivity	Natural gas pipeline data set	Supervised
Paper[42]	KNN	accuracy rate	KDD99	Supervised
Paper[44]	Logistic regression	Sensitivity and error rate	KDD-cup 1999	Supervised
Paper[47]	Decision tree	Accuracy, precision, etc.	ISCXIDS2012 , NSL-KDD	Supervised
Paper[71]	OCSVM	accuracy rate	Data flow	Unsupervised
Paper[72]	OCSVM	Accuracy, precision, recall and false alarm rate	Unpublished data set	Unsupervised
Paper[75]	K-means	RMSE, execution time	Data flow	Unsupervised
Paper[107]	SVM, random forest	Accuracy, recall, precision, F1	Natural gas pipeline data	Unsupervised
Paper[54]	Semi-supervision and information gain	Accuracy, detection rate, false alarm rate, false alarm rate	Natural gas pipeline data set	Semi-supervision
Paper[56]	Semi-random SVM and semi-random forest	Accuracy, precision, recall, F1 value	NSL-KDD	Semi-supervision
Paper[57]	Rocchio, LIBLINEAR	Accuracy, recall, F1 value	KDD99	Semi-supervision

C. Intrusion Detection Algorithm Based on Semi-supervised Learning

Semi-supervised learning method^[48] exploring hidden features of a large number of unlabeled data through a small amount of labeled data is a learning method between supervised and unsupervised, which has become a research hot spot in machine learning^[51].

At present, there are many industrial control intrusion detection algorithms based on semi-supervision. For example, a semi-supervised industrial control intrusion detection algorithm based on the improved K-means algorithm and information gain is constructed in [54]. The algorithm has achieved good detection results on data sets of gas transmission system and water storage system^[53]. A semi-supervised industrial control intrusion detection method based on K-means of incomplete information is implemented. This method can mine known types of attacks by detecting network traffic data, which can improve the detection accuracy of abnormal behaviors. Experiments show that the semi-supervised random

forest method has better detection effect than the semi-random SVM method in network intrusion detection in paper [56]. Paper [57] Combines Rocchio and LIBLINEAR[58] and implements a semi-supervised industrial control intrusion detection algorithm.

Industrial control systems have high requirements for real-time. In order to improve the real-time performance of industrial control intrusion detection algorithms, [105] combines machine learning algorithm with statistical algorithm and proposes a high real-time industrial control intrusion detection model. This model is not only with high detection accuracy, but also with low computational complexity. It is suitable for the resource-constrained situation in ICS, and can effectively detect various network attacks, malicious manipulation and other abnormal events in real industrial control scenarios.

Although machine learning algorithms have been widely used in many fields for the excellent generalization ability. There are still new challenges to industrial control intrusion

detection based on machine learning for the increasingly huge, nonlinear, high-dimensional and noisy data together with the development of intelligent and integrated ICS.

IV. INTRUSION DETECTION METHOD BASED ON DEEP

Although the industrial control intrusion detection algorithm based on machine learning can achieve high detection accuracy, it needs industry experts to manually extract data features before detection, which is a huge workload and takes a long time. With the rapid development of artificial intelligence, the application of deep learning is more and more extensive. Deep neural network has a strong ability to extract data features, and intrusion detection algorithm based on deep learning can avoid a lot of artificial feature extraction. Many researchers begin to study intrusion detection based on deep learning. Paper [79] comprehensively summarizes the intrusion detection methods based on deep learning in recent years. At present, the most influential intrusion detection methods based on deep learning mainly include the design of DeepLog framework for log anomaly detection based on LSTM neural network in [80]. DeepLog framework can not only identify normal log data, but also be used for online anomaly detection of incoming log data in streaming mode. In [81], a multi-modal intrusion detection system based on LSTM is proposed. In [82], NLP is introduced into intrusion detection system, and a NeuralLog framework is put forward, which extracts the original log information into semantic vectors, and then classifies these semantic vectors by multi-Transformer model to identify the abnormal information.

Although deep learning has been widely and deeply applied in the field of intrusion detection, some complex deep neural network models are difficult to apply to ICS intrusion detection because of the limited computing resources and storage resources of industrial control systems. Therefore, when researching industrial control intrusion detection based on deep neural network, we need to focus on the complexity of the model, and make the network structure as simple as possible while ensuring the detection accuracy. Therefore, this paper studies the intrusion detection algorithms of industrial control based on depth in recent years, and the summary is shown in Table 2

A. Intrusion Detection Algorithm Based on Sequential Neural Network

The data in industrial control system has strong time series, so the time series neural network which is good at processing serial data can be used to design the intrusion detection algorithm for industrial control. Commonly used sequential neural networks include RNN(Recurrent Neural Network), LSTM(Long Short-Term Memory) and GRU(Gate Recurrent Unit). RNN is composed of a series of neurons that need serial calculation, and the output of the former neuron will be the input of the latter neuron, which makes RNN have strong timing and is suitable for processing ICS data. For example, [83] describes how to treat traffic packets as a kind of serial data and process these serial data through RNN network. The process of ICS intrusion detection based on RNN is shown in Figure 3, which shows the output data, the input data, and the weights of U, V and W.

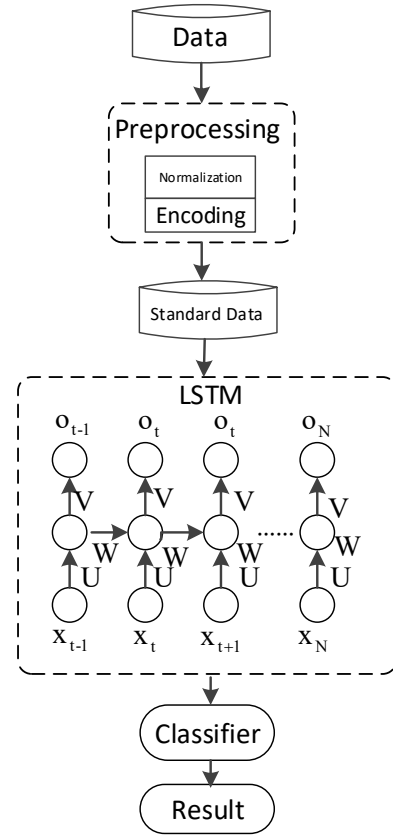


Fig. 3. Intrusion detection model based on RNN.

Although RNN can be used to process the sequential data of industrial control systems, RNN can't establish long dependence relationship, and a series of problems such as gradient disappearance (or explosion) often occur in the process of training. Therefore, many researchers have introduced LSTM into industrial control intrusion detection systems.

In order to solve the problems existing in RNN, LSTM adds control units such as Input Gate, Output Gate and Forget Gate on the basis of RNN. LSTM can realize the long dependence of sequence data and is an optimization of RNN. By combining signature packet-based detection methods with LSTM time series detection methods, an intrusion detection model is constructed in [84]. In this model, some abnormal traffic is detected by Bloom Filter detector, and then another abnormal traffic is detected by LSTM. Yu bangbing^[85] have also proved that LSTM has better detection effect than RNN in the field of industrial control intrusion detection through experiments in the field of natural gas control system.

In addition, GRU, as a variant of LSTM, can also be used in industrial control intrusion detection system. Xu et al.[86] proposes a misuse detection model based on BigRu (Bidirectional Gate Recurrent Unit). This detection model can keep long sequence information of industrial control data, and to some extent, it can avoid the problems of gradient disappearance (or explosion).

TABLE II. SUMMARY OF THE INTRUSION DETECTION METHODS FOR INDUSTRIAL CONTROL SYSTEM BASED ON DL

Papers	Deep learning algorithm	evaluating indicator	Or scene data set
Paper [67]	RNN	False alarm rate	SWaT
Paper [78]	eSNN	Accuracy, precision, recall, F1 value, RMSE and ORC	Natural gas pipeline data set
Paper [84]	LSTM	False alarm rate, accuracy rate, recall rate, precision rate and F1	Natural gas pipeline data set
Paper [85]	LSTM	Accuracy, false alarm rate, false alarm rate	Natural gas pipeline data set
Paper [86]	BiGRU	Accuracy, DR, FPR	KD99 and NSL-KDD
Paper [89]	CNN	Accuracy and false detection rate	KD99
Paper [90]	CNN+BiLSTM	Loss, recognition rate	Metallurgical industry control system
Paper [91]	CNN-BiLSTM	Accuracy, false alarm rate, false alarm rate	Natural gas pipeline data set
Paper [93]	AE	Accuracy, precision, recall, F1 value	BATADAL
Paper [94]	LSTM-Autoencoder	Accuracy, precision, recall, F1 value, RMSE	Unpublished data set
Paper [96]	Spares AE	Accuracy, precision and recall rate	NSL-KDD
Paper [97]	DAE	accuracy rate	KDD99
Paper [98]	NDAE	accuracy rate	KDD99
Paper [99]	SDAE+AE	Accuracy, precision and recall rate	KDD99
Paper [100]	CDBN	Detection rate, ROC	FDI attack on power grid
Paper [101]	CUSUM and Bad-Data		SWaT, WADI
Paper [102]	Behavior Modeling Based on Fuzzy Rules (T1-FLSs)	Accuracy, false positive rate, false true rate	Analog data of embedded hardware system
Paper [103]	Behavior Modeling Based on Fuzzy Rules (T2-FL)	Accuracy, false positive rate, false true rate	Analog data of embedded hardware system
Paper [104]	Behavior Modeling Based on Fuzzy Rules (IT2-FLS)	Accuracy, false positive rate, false true rate	Analog data of embedded hardware system
Paper [106]	CNN	Accuracy, precision, recall, F1 value	Unpublished data set

B. Intrusion Detection Algorithm Based on Convolutional Neural Network

Convolutional Neural Network (CNN) refers to a kind of neural network that performs convolution operation through convolution kernel. The composition of CNN can be divided into input layer, convolution layer, pool layer, full connection layer and output layer. CNN has a strong ability to extract local features, and was originally applied in the image field. CNN has strong generalization ability and can be applied to various classification tasks. At present, it has been widely used in the field of intrusion detection. [87] takes the data packet sequence as input, the data stream is preprocessed into image form, and classified by CNN network. Xiao et al. [108] proposes an intrusion detection model based on feature dimensionality reduction and CNN. This model first reduces the dimensionality of data, and then converts the data after dimensionality reduction into a grayscale image, which is input into CNN network for feature extraction and classification. Khan et al. [88] improves CNN in the intrusion detection model. The improved CNN has three hidden layers, and each hidden layer contains a convolution layer and a pool layer. In these three hidden layers, the number of convolution kernels increases in turn. This method can map the input low latitude data into high latitude data, thus enhancing the ability of extracting data features of the model.

For industrial control intrusion detection based on CNN, the data in ICS can be preprocessed into the form of image data, and then digital features can be extracted and classified by convolution operation in CNN. The process structure is shown in Figure 4.

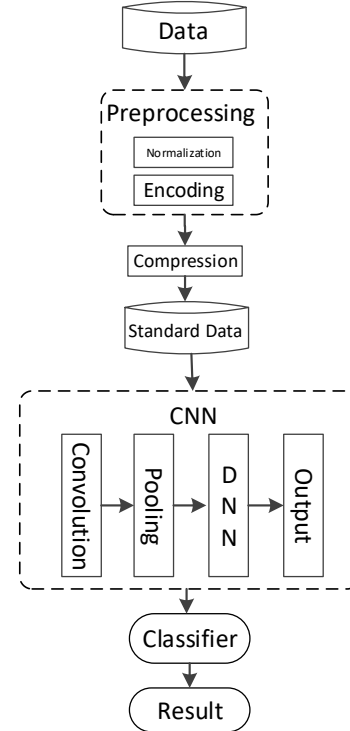


Fig. 4. Intrusion detection model based on CNN.

With the feature of weight sharing of CNN, it can effectively reduce the training parameters and computing resources in the training process, and well meet the resource limitation of industrial control system. Based on CNN network, Zhao Zhiyang et al. [89] put forward an intrusion detection

algorithm for the security of power grid control system, which optimizes the network structure by cascading layers. Although the intrusion detection algorithm based on CNN network can avoid the problems of gradient disappearance or gradient explosion, its ability to learn features in time dimension is not as good as that of LSTM. Therefore, many researchers will combine LSTM with CNN network when implementing intrusion detection algorithm. For example, Song Yu et al.^[90] combines BiLSTM with CNN in their intrusion detection algorithm for ARP attack in ICS. [91] proposes an intrusion detection model combining correlation information entropy, CNN and BiLSTM, which can reduce the dimension of data. In this model, firstly, the features are selected by the relevant information entropy, and then the feature dimensions are reduced, and then the classification is carried out by CNN-BiLSTM.

C. Intrusion Detection Algorithm Based on Self-encoder

AutoEncoder (AE)^[92] is an unsupervised deep learning algorithm. Its purpose is to re-encode the input data and make the input data close to the input data. This makes the low-dimensional neurons in the deep hidden layer replace the high-latitude neurons in the shallow hidden layer, thus reducing the data dimension. In the process of encoding and decoding, the self-encoder can learn the inherent feature distribution of data, so that the data can be classified. Therefore, the data of industrial control system can be classified by self-encoder, and whether the industrial control data is abnormal or not can be judged. The architecture of industrial control intrusion detection model based on self-encoder is shown in Figure 5.

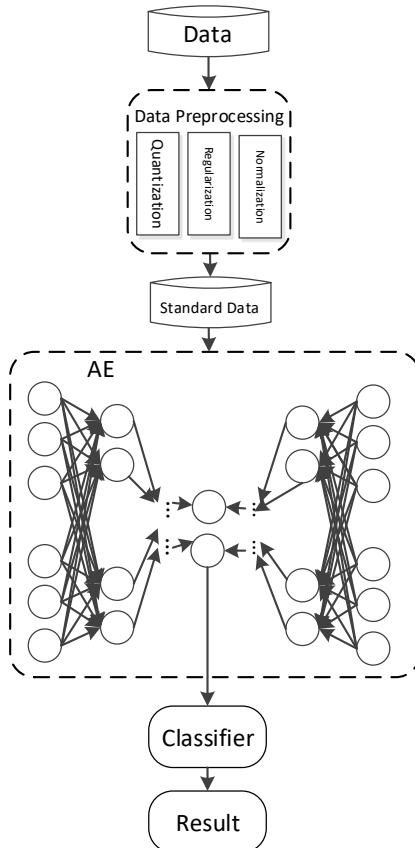


Fig. 5. Intrusion detection model based on self-encoder.

At present, self-encoder has been widely used in the field of intrusion detection. For example, Gurung et al.^[96] propose an intrusion detection model based on sparse self-encoder, which achieves high accuracy on NSL-KDD data sets. Farahnakian et al.^[97] connect four self-encoders and established an intrusion detection model based on depth self-encoder. Shone et al.^[98] even establish an intrusion detection model by combining the self-compiled encoder of asymmetric multi-hidden layers with random variables. This model can reduce the resources needed for calculation, and can well meet the characteristics of resource first in industrial control system. Wang et al.^[99] combine SDAE (Stacked Denoising Auto Encoder) with ELM, and propose an intrusion detection model with good generalization ability, which can satisfy the feature that industrial control system can't stop. In [94], an intrusion detection model based on LSTM-AutoEncoder is proposed. In this model, the bad data and missing data in the data are detected by LSTM, and these data are eliminated, and then whether there is any abnormality in the remaining data is judged by AE.

Besides these common neural networks, some new neural networks have been applied in the field of industrial control intrusion detection. Demertzis et al.^[78] propose an intrusion detection method SOCCADF based on evolving Spiking Neural Network (eSNN). SOCCADF algorithm can only be trained by positive samples, which is an unsupervised algorithm. Then, based on CDBN (Conditional Deep Belief Network), an intrusion detection algorithm for false data injection attacks in smart grid is proposed by He et al.^[100]. The algorithm performs well in different environmental noise experiments. Rassam^[109] and Siphesihle^[113] do the research on intrusion detection based on the Artificial Immune Network. Kavitha et al.^[110] embedded the time behavior of attacks into TDNN neural network and proposed a host based intrusion detection system

ICS has the characteristics of real-time requirement and difficulty to pause, so in the field of industrial control intrusion detection, it is necessary to take detection delay as an important evaluation index. In [101], the statistics-based detection technology (CUSUM and Bad-Data) and the deep learning-based detection technology (NoisePrint) are compared, and it is found that the deep learning-based detection technology can detect advanced attack modes, but its delay is large which makes it unfit for the industrial control scenes with real-time requirements. In order to reduce the cost of neural network algorithm, an intrusion detection method for industrial control based on fuzzy rules modeling behavior is proposed in [102]. This method will construct a fuzzy logic rule base for normal behavior. In order to reduce the false alarm rate, Kavitha et al.^[109] applied three Intuitionistic fuzzy rules to the intrusion detection algorithm. Gunay et al.^[114] proposed a method to dynamically detect Statistic Approached Dynamically Detecting Security Threats and Updating a Signature-Based Intrusion Detection System's Database in NGN.

V. CONCLUDING REMARKS

With the development of ICS intelligence and information technology, ICS is increasingly connected with the Internet, which further increases the security threats faced by ICS. Many

ICS are the key production facilities of the country. Once attacked, they will not only bring huge economic losses, but also pose a serious threat to the national security. Industrial control intrusion detection technology has gradually become one of the important technologies to protect ICS security.

- With the development of artificial intelligence, many researchers began to study industrial control intrusion detection based on machine learning and deep learning. The detection method based on machine learning has high detection accuracy, but the intrusion detection algorithm based on machine learning needs data reduction and artificial feature extraction when faced with high-dimensional industrial control data.
- The detection method based on deep learning has strong feature extraction ability, but the algorithm based on deep learning usually has too complex network structure, huge parameters and high computational complexity, so it is difficult to apply in ICS with limited resources.
- ICS has the characteristics of real-time requirement, limited resources, high data dimension and much noise data. How to study industrial control intrusion detection and improve the security of ICS according to these characteristics of ICS is the key research direction in the future.

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